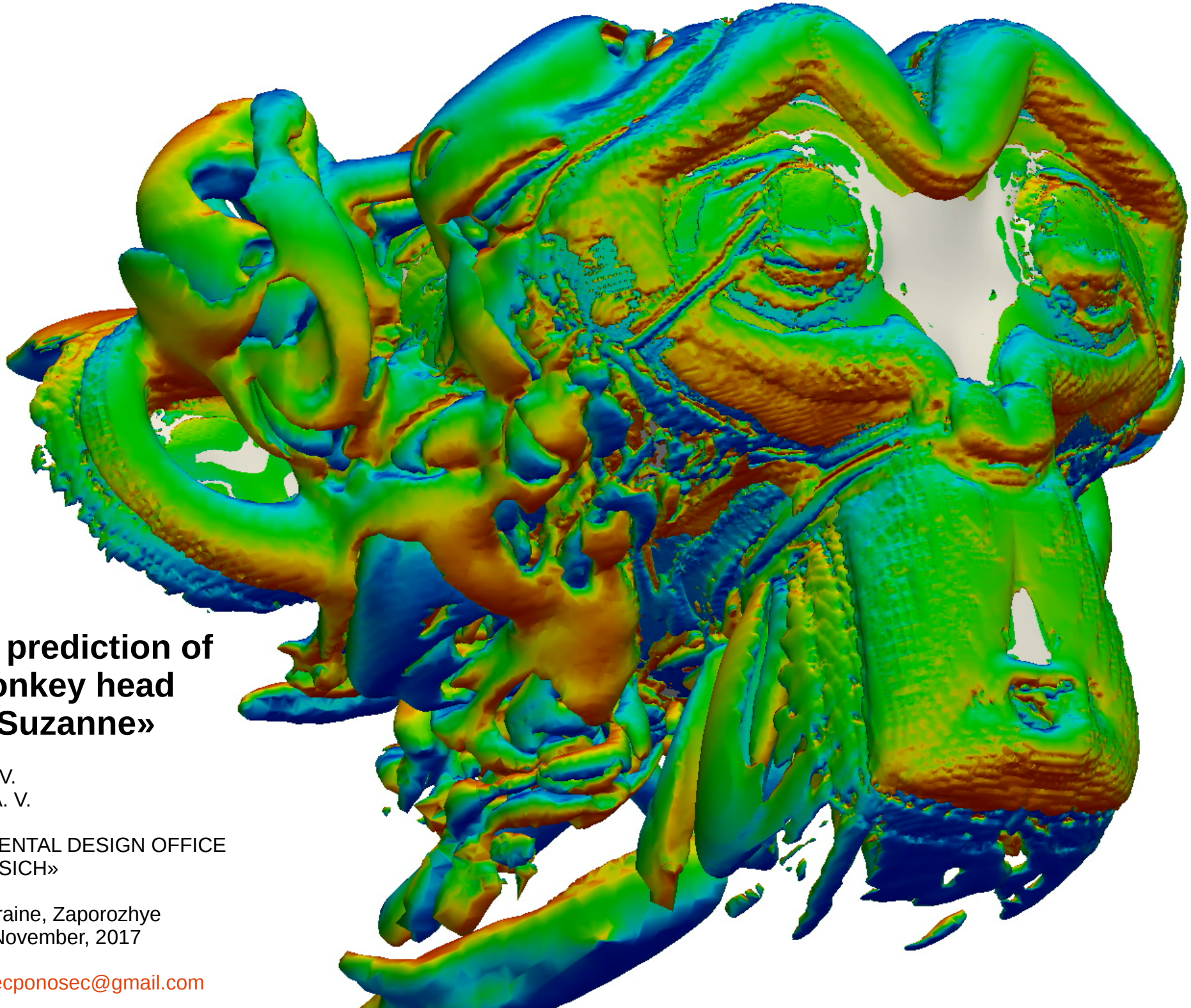


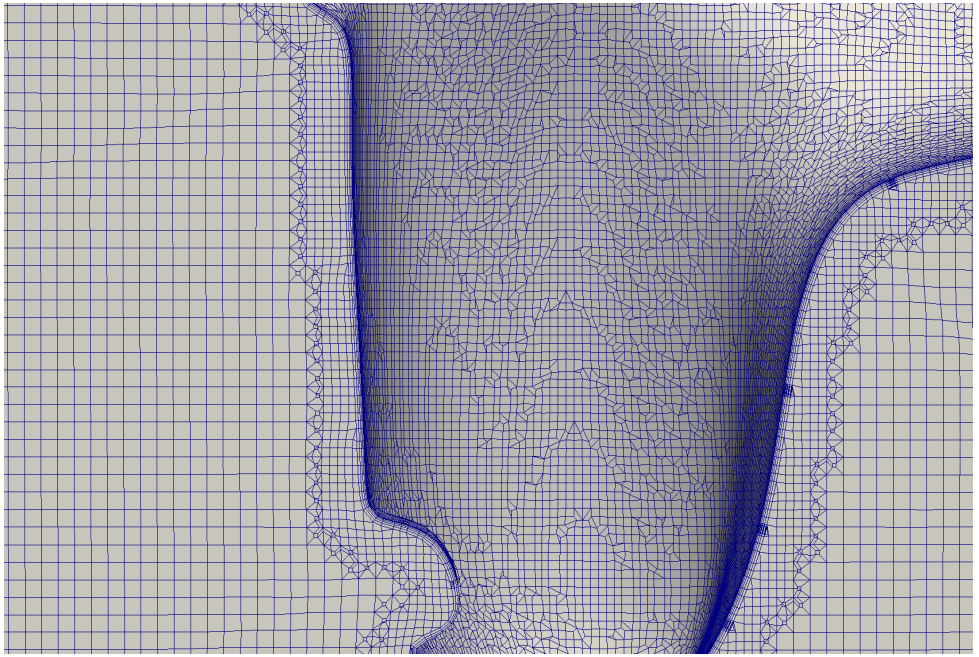
Drag prediction of monkey head «Suzanne»

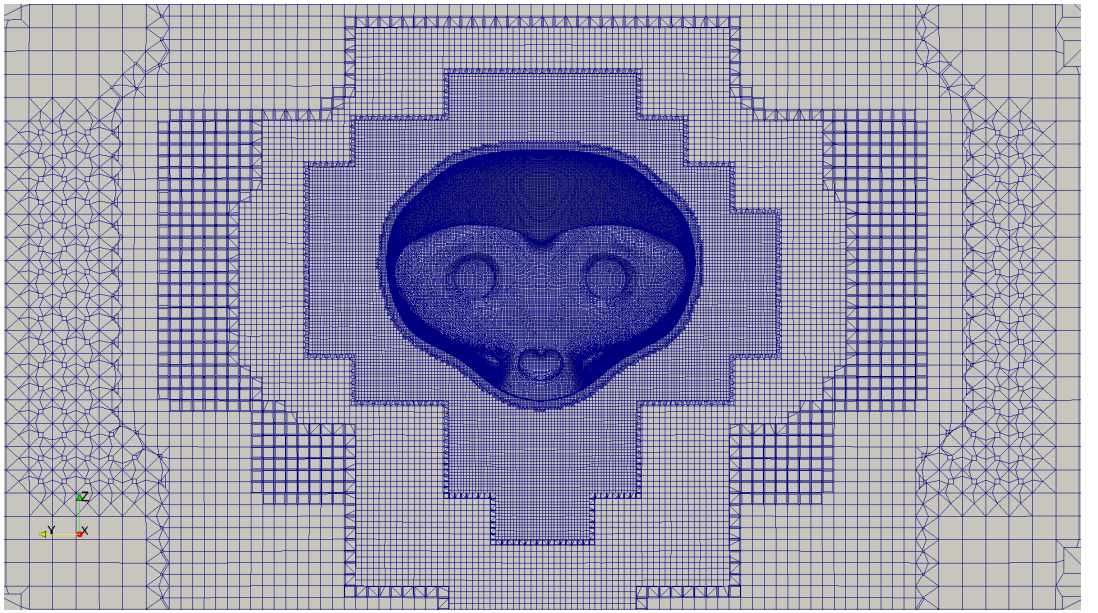
Prytula A. V.
Lebedko A. V.

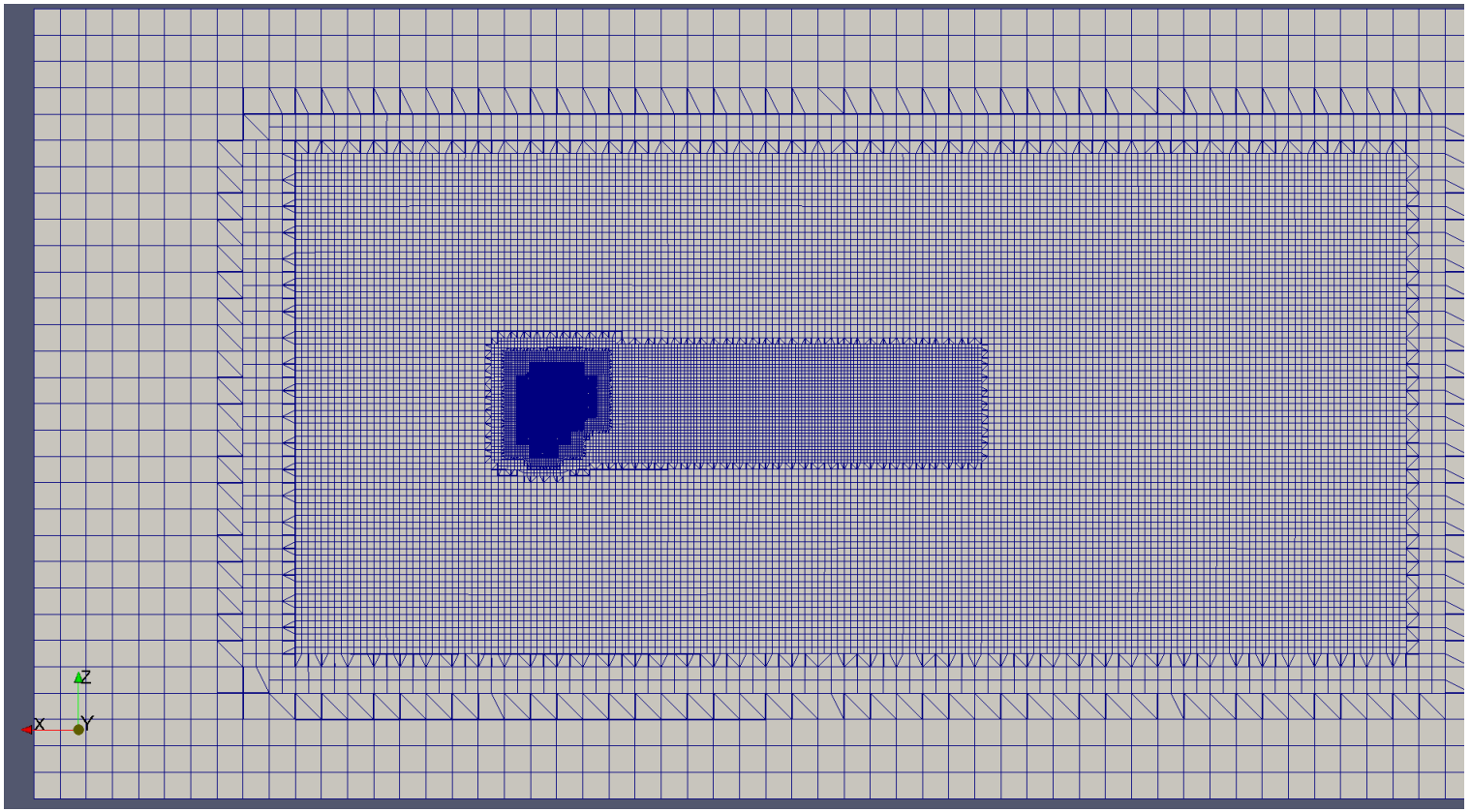
EXPERIMENTAL DESIGN OFFICE
«MOTOR SICH»

Ukraine, Zaporozhye
November, 2017

bronenosecponosec@gmail.com







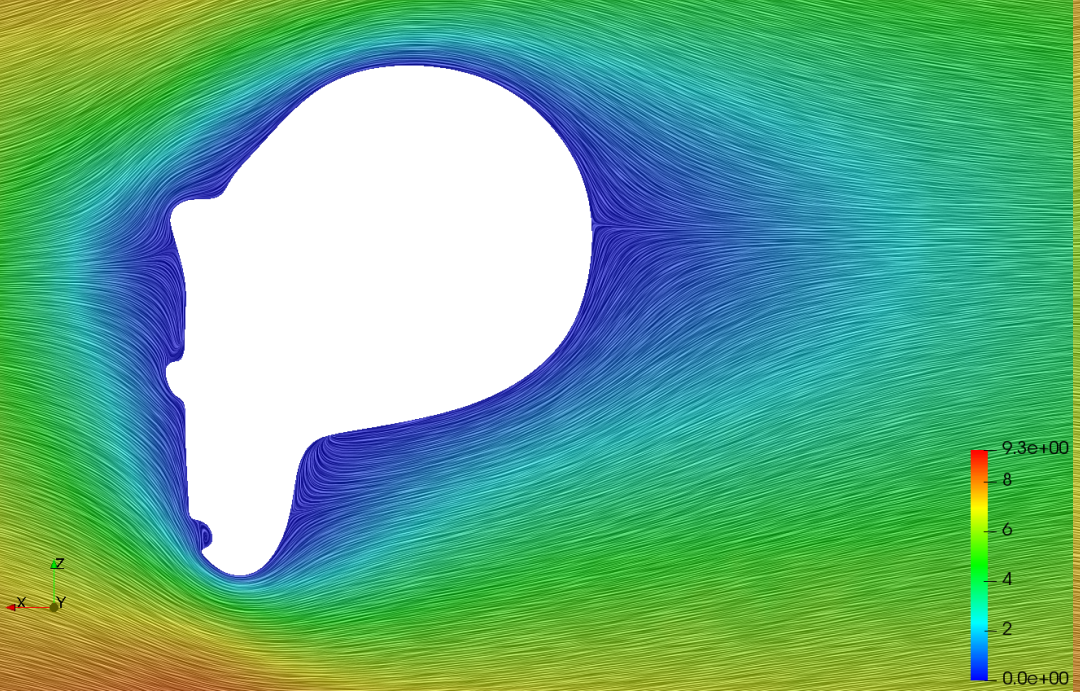
Mesh generated with snappyHexMesh:

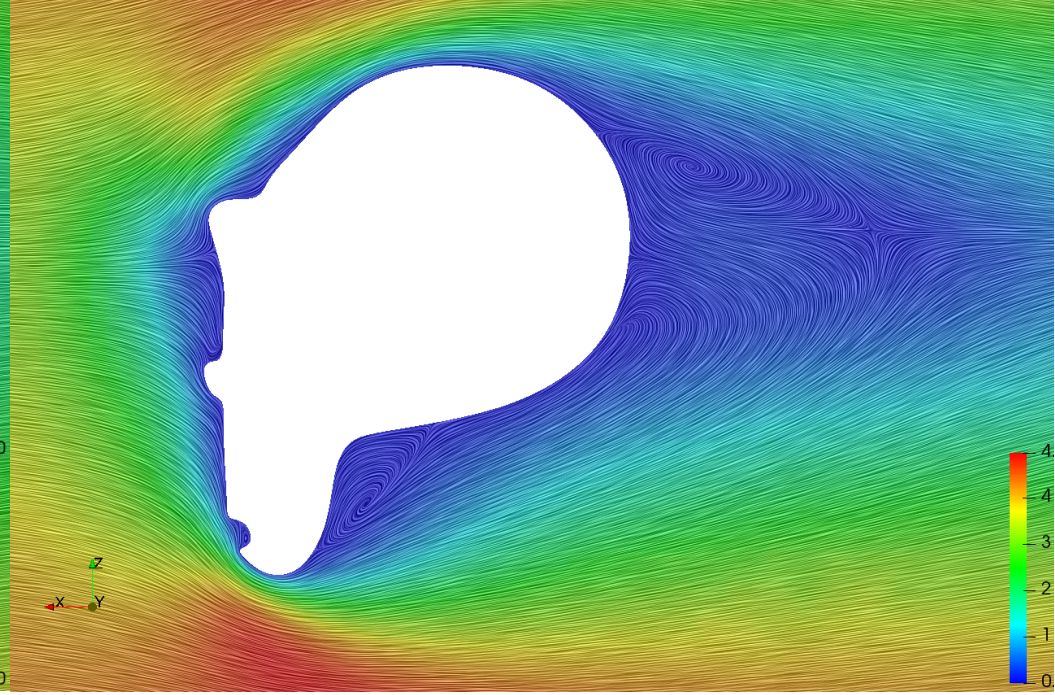
- blockMesh 60x30x30 m^3
- 2 refinement regions
- 8 levels of refinement
- 3 boundary layers
- 98% layers coverage
- max cell skewness 2.2
- 4 599 272 cells

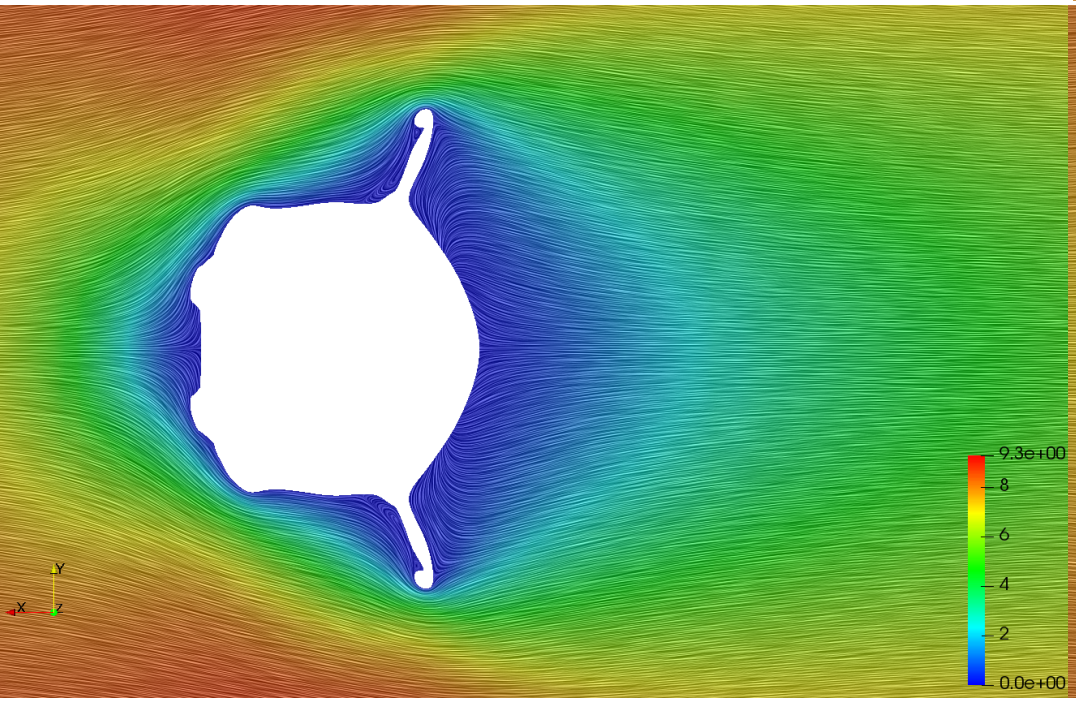
Solver: simpleFoam

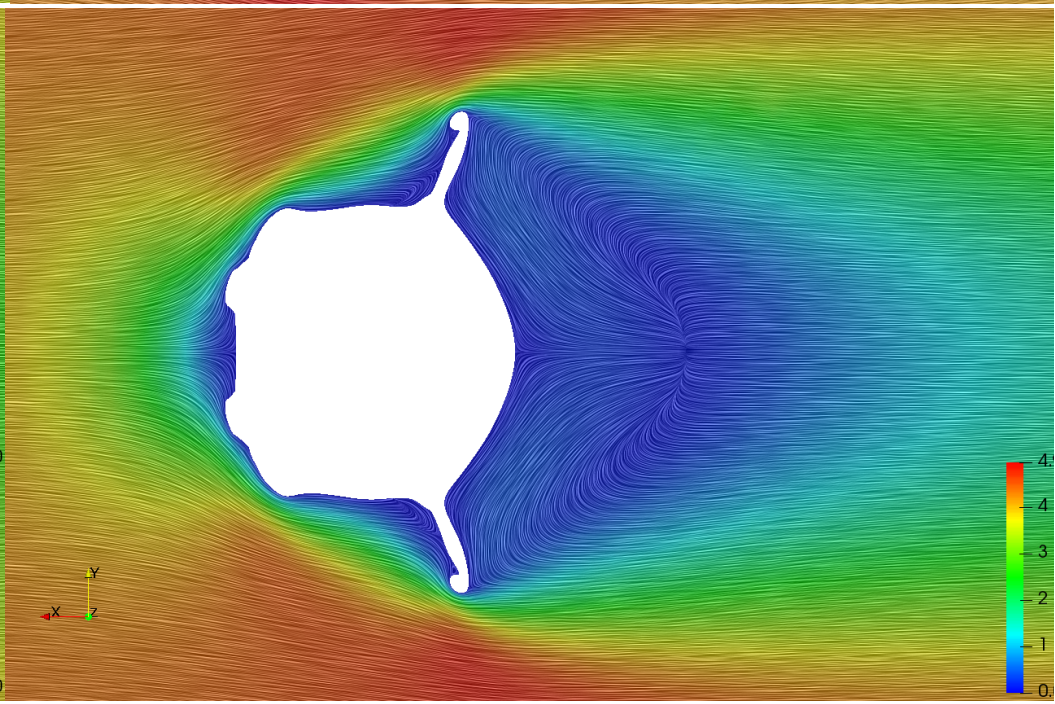
simulationType RAS
RASModel kOmegaSST

Processor: Intel Xeon W3550
RAM: 12 GB



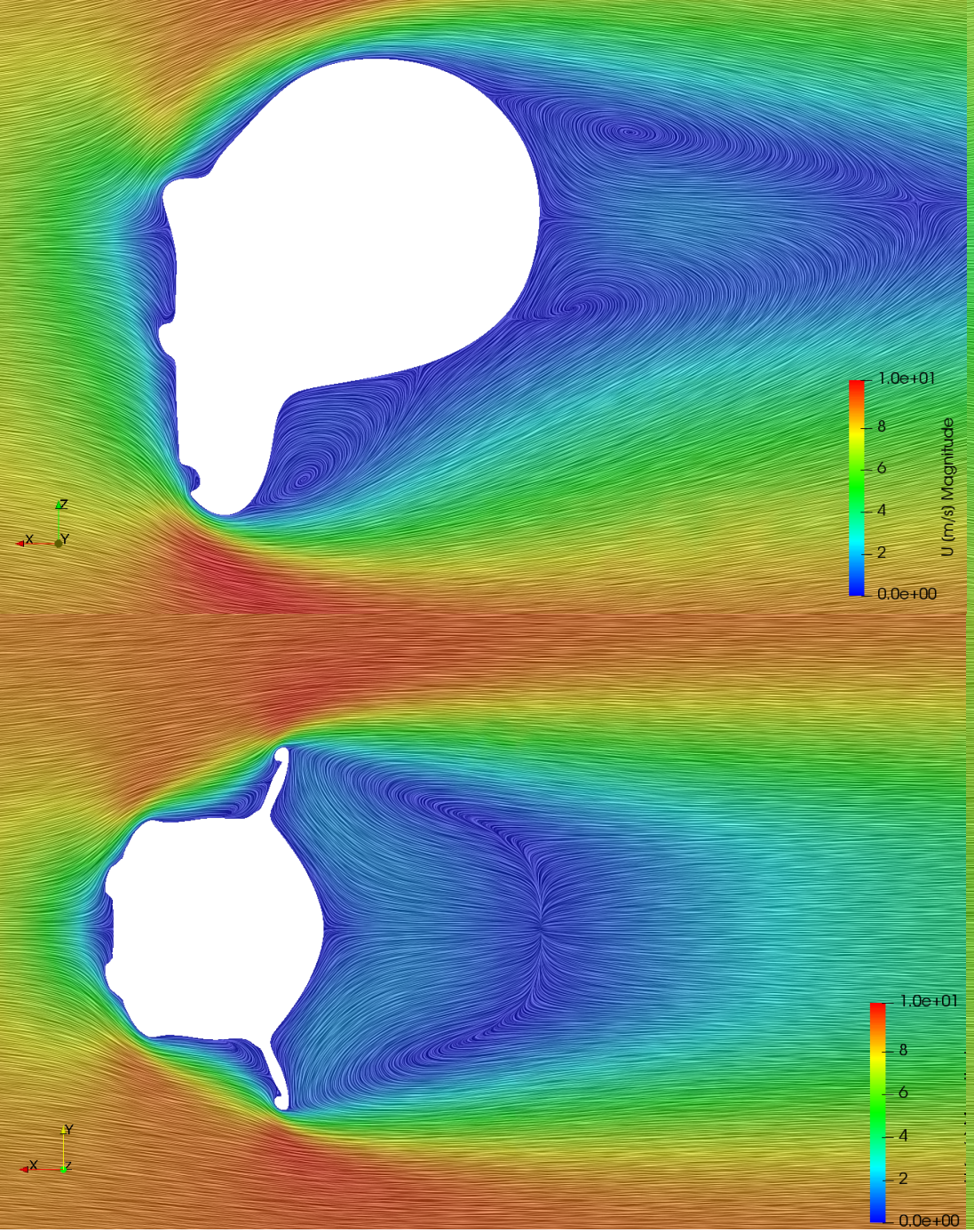


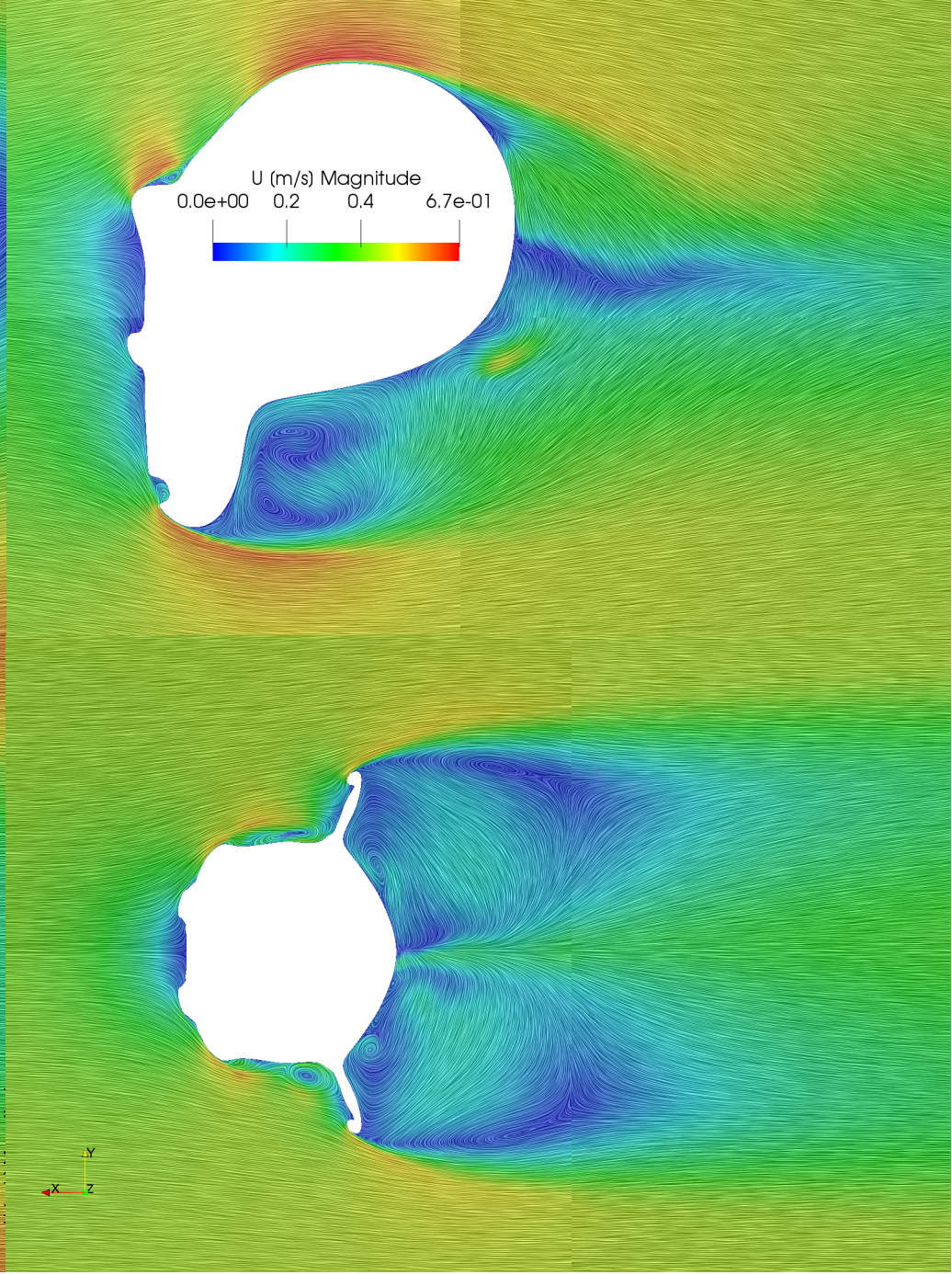




Re = 10
Cd = 4.35

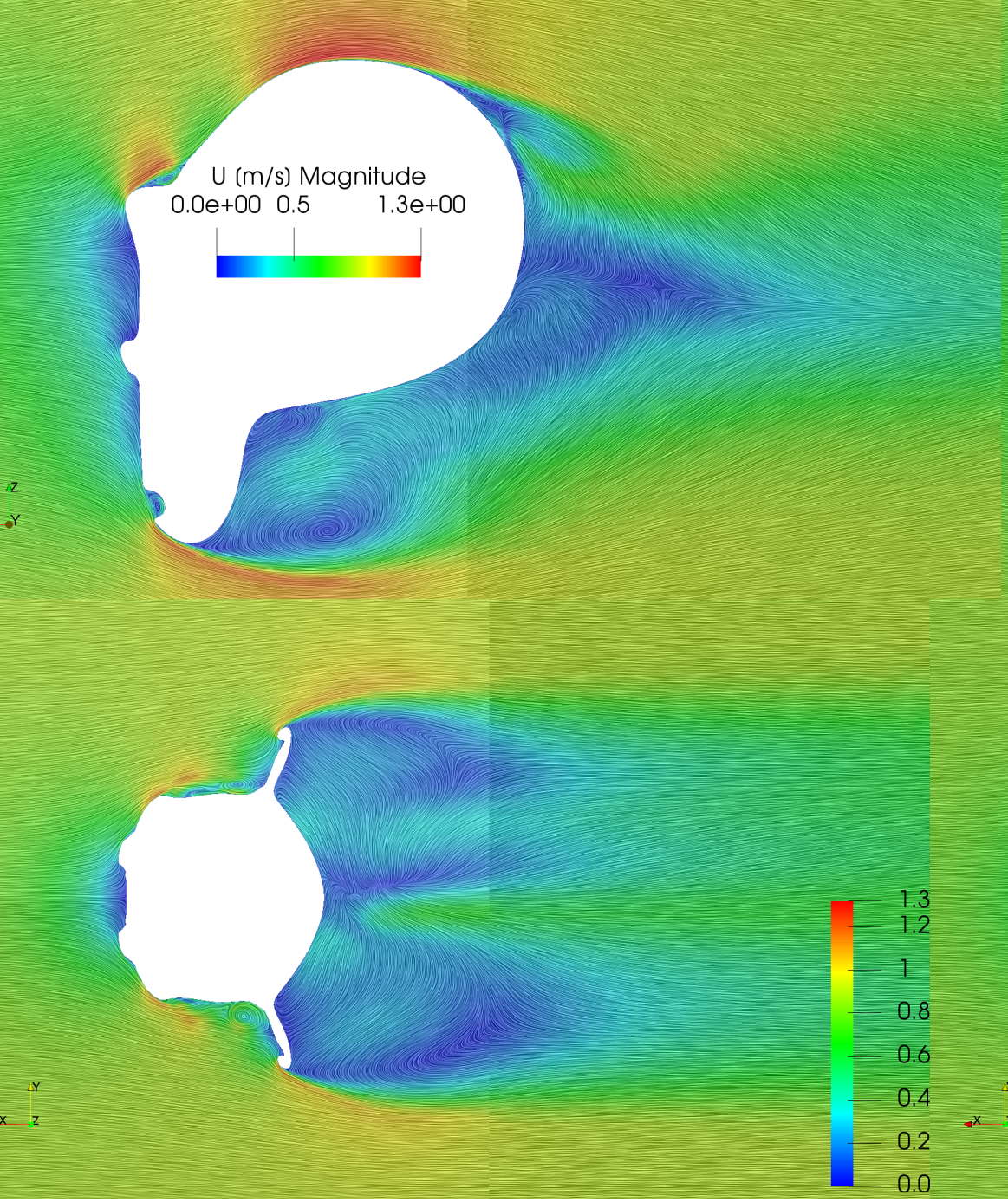
Re = 50
Cd = 1.68



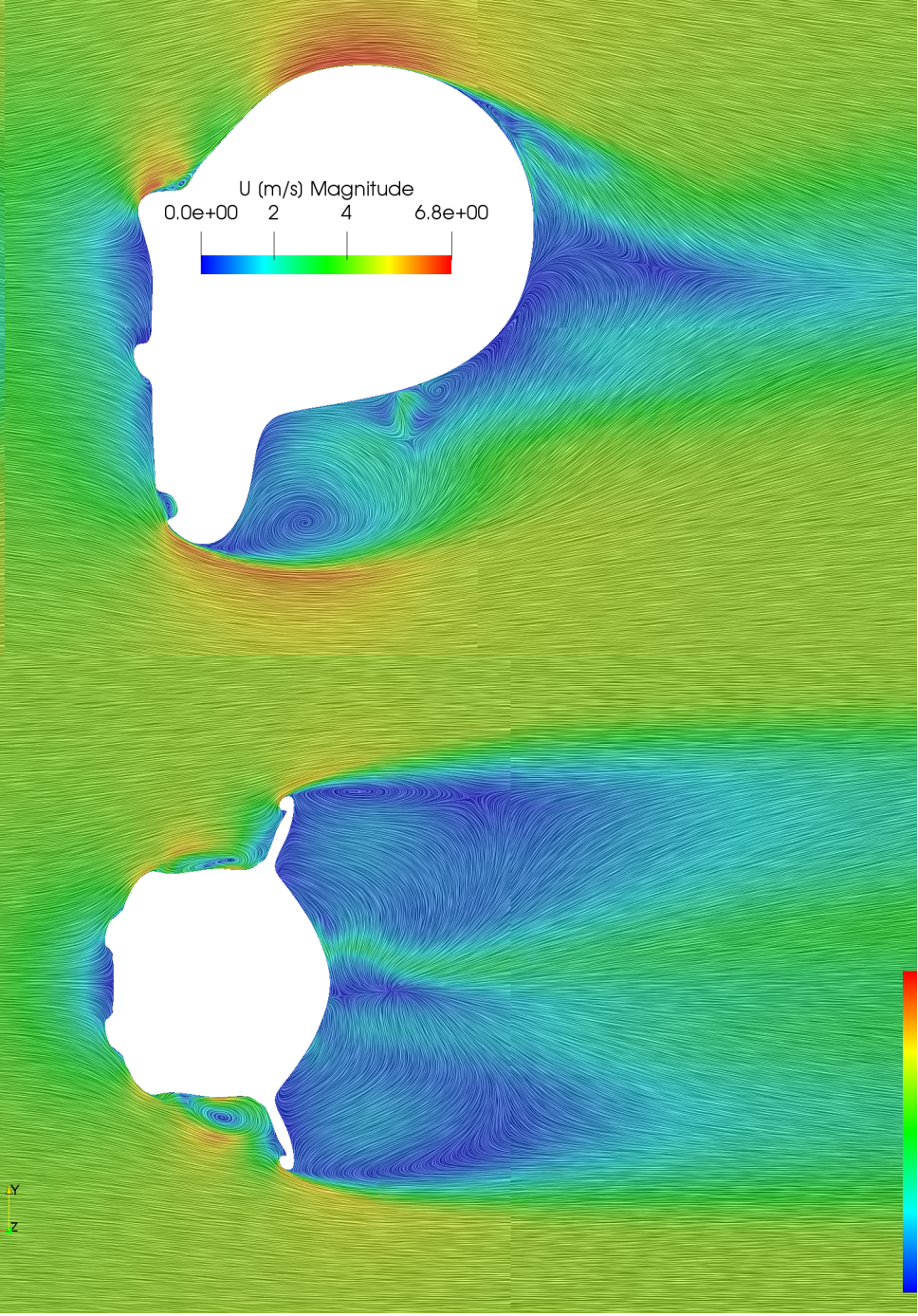


Re = 100
Cd = 1.22

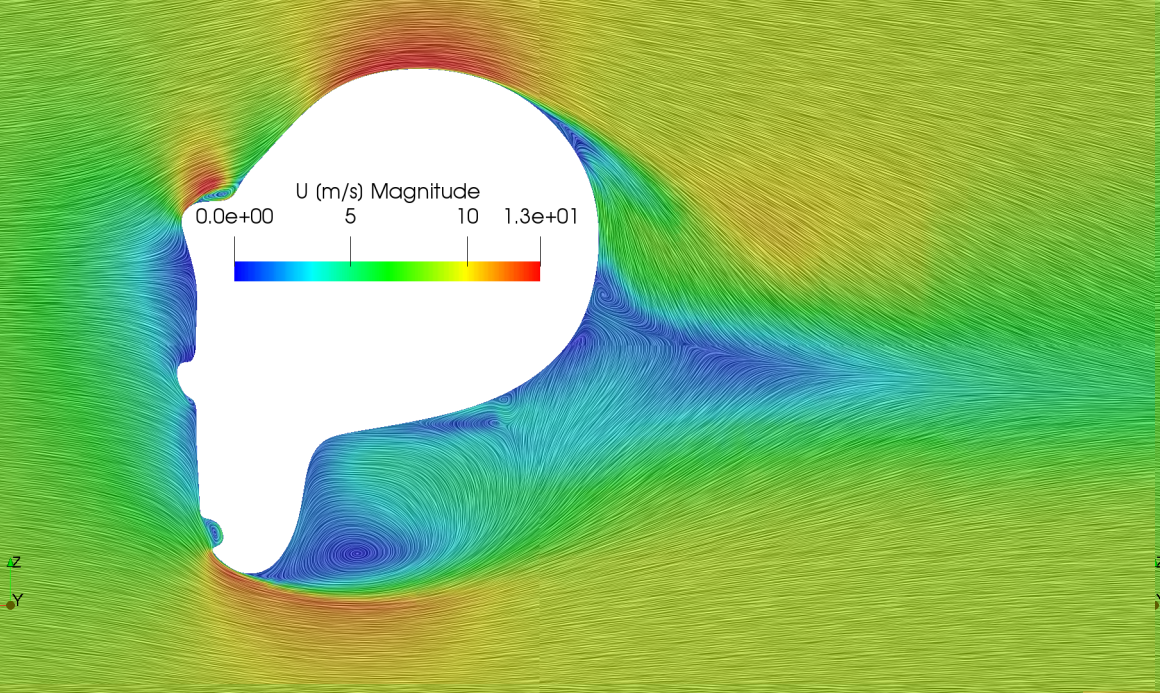
Re = 50 000
Cd = 0.73

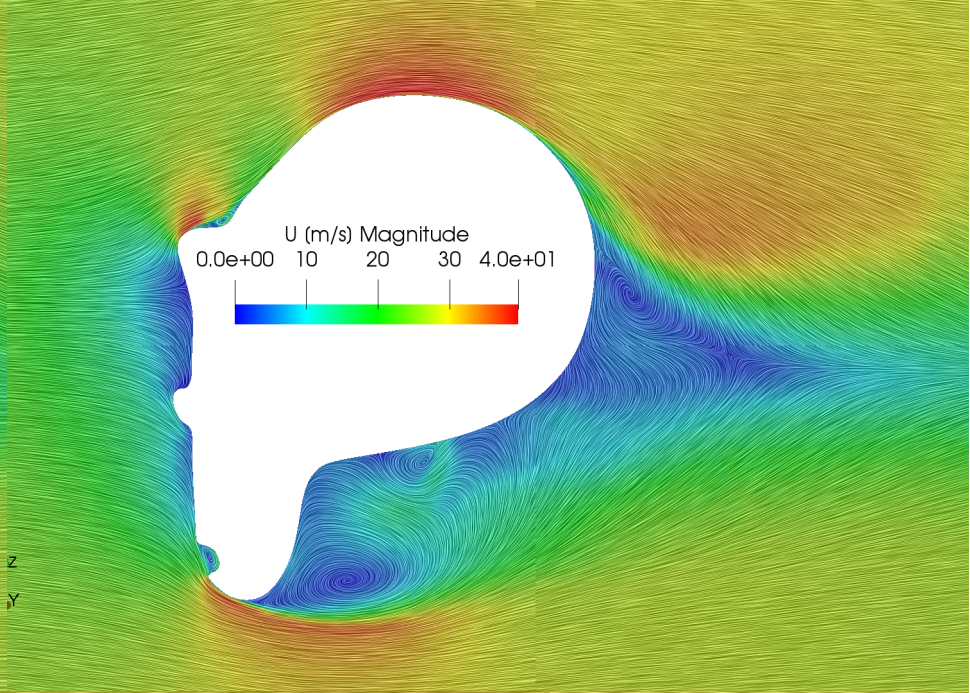


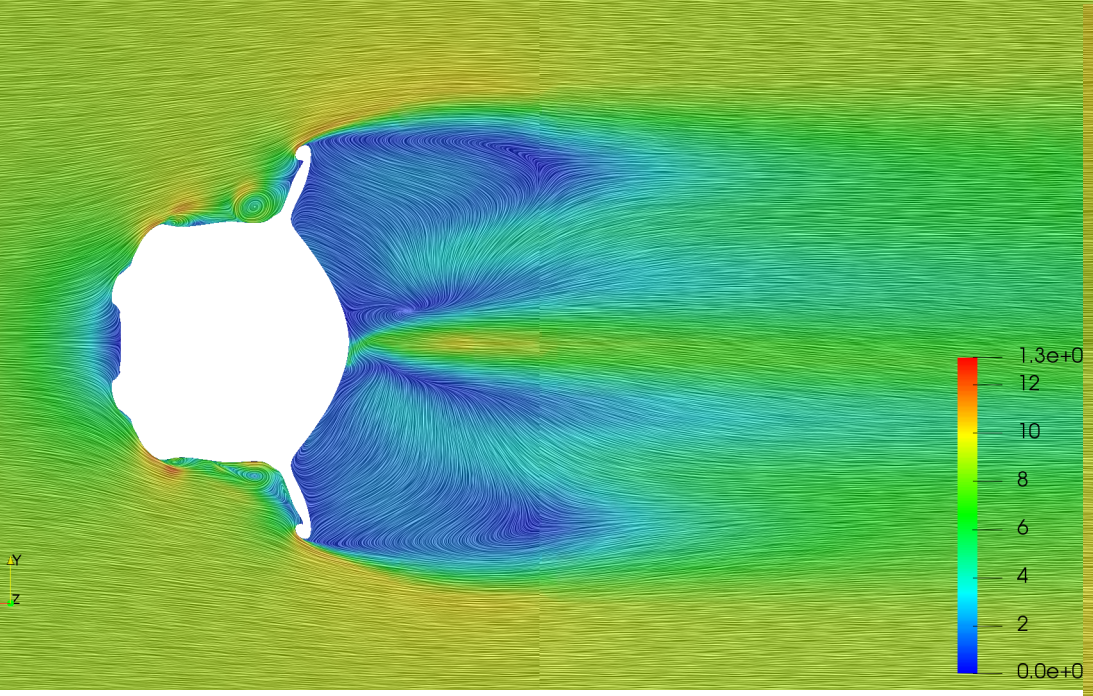
Re = 100 000
Cd = 0.71

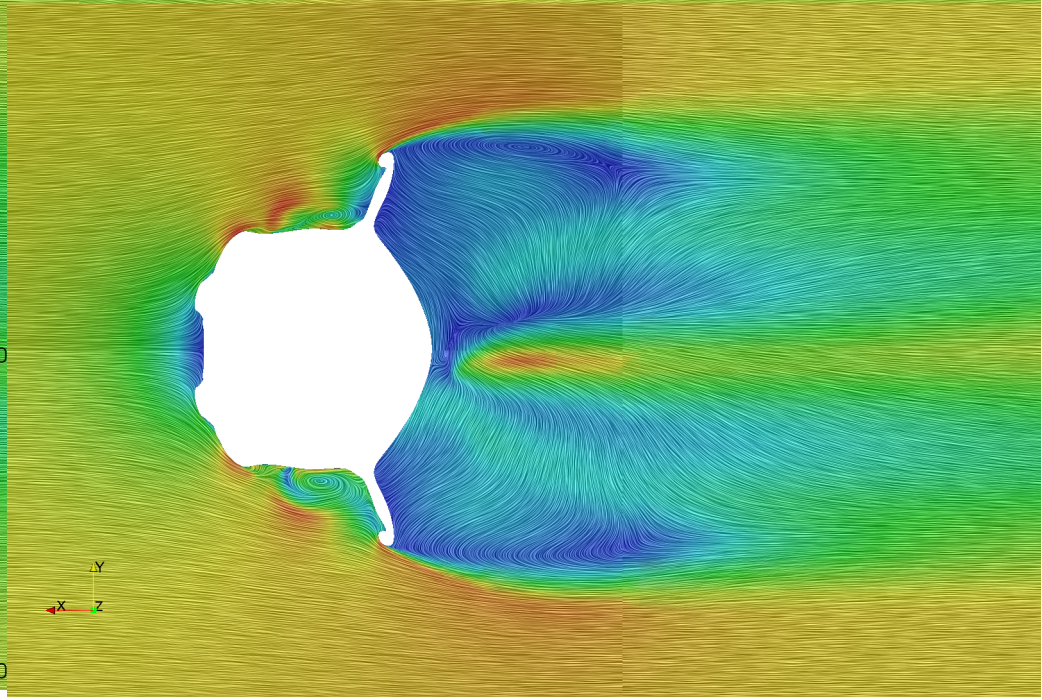


Re = 500 000
Cd = 0.67



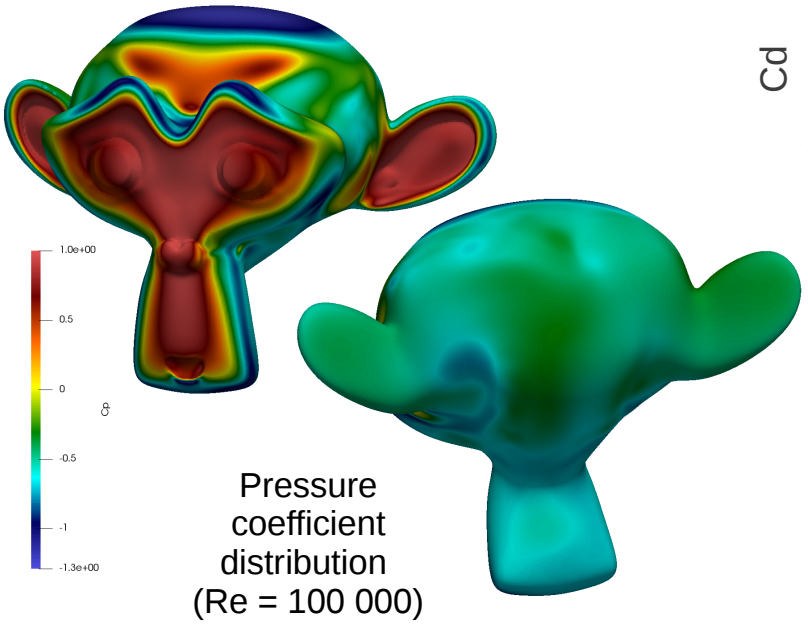




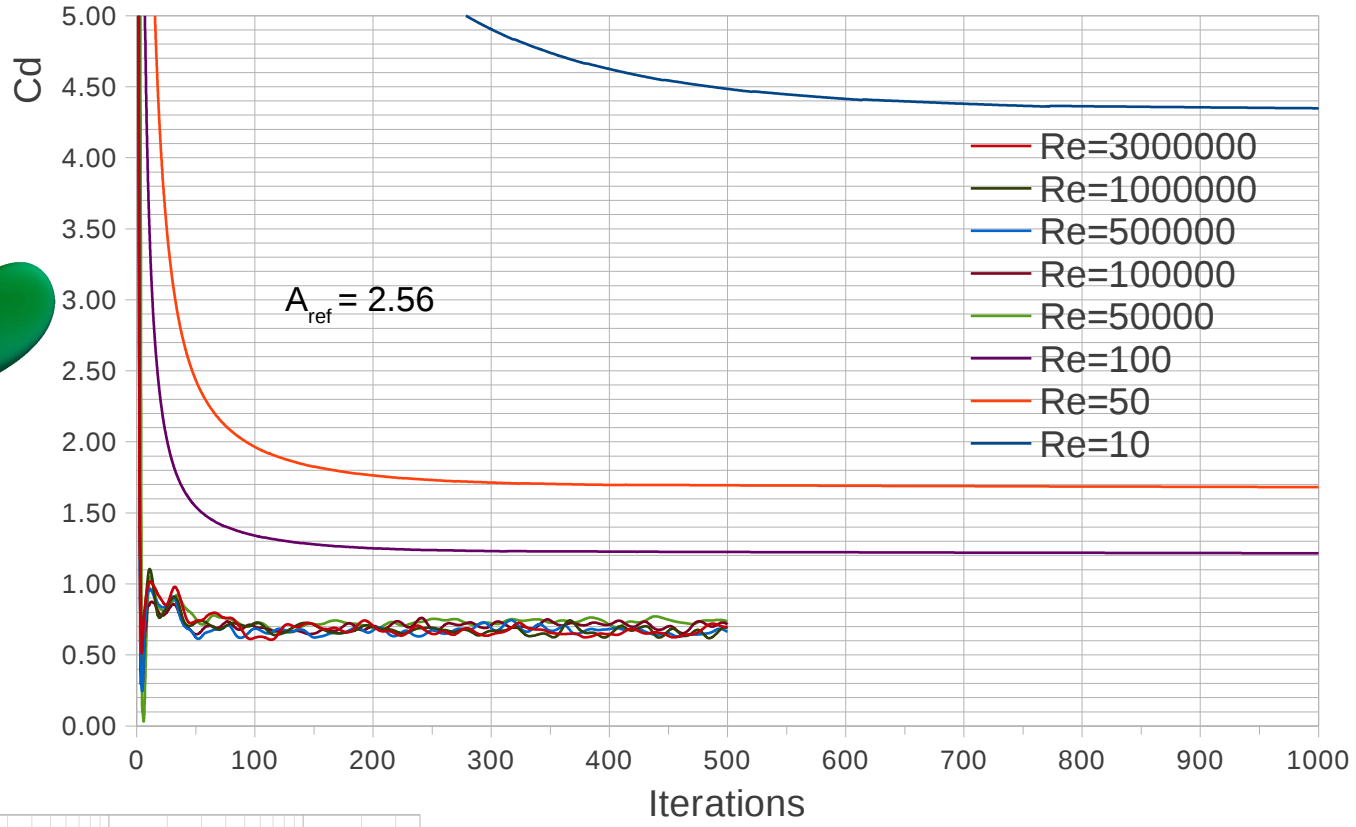


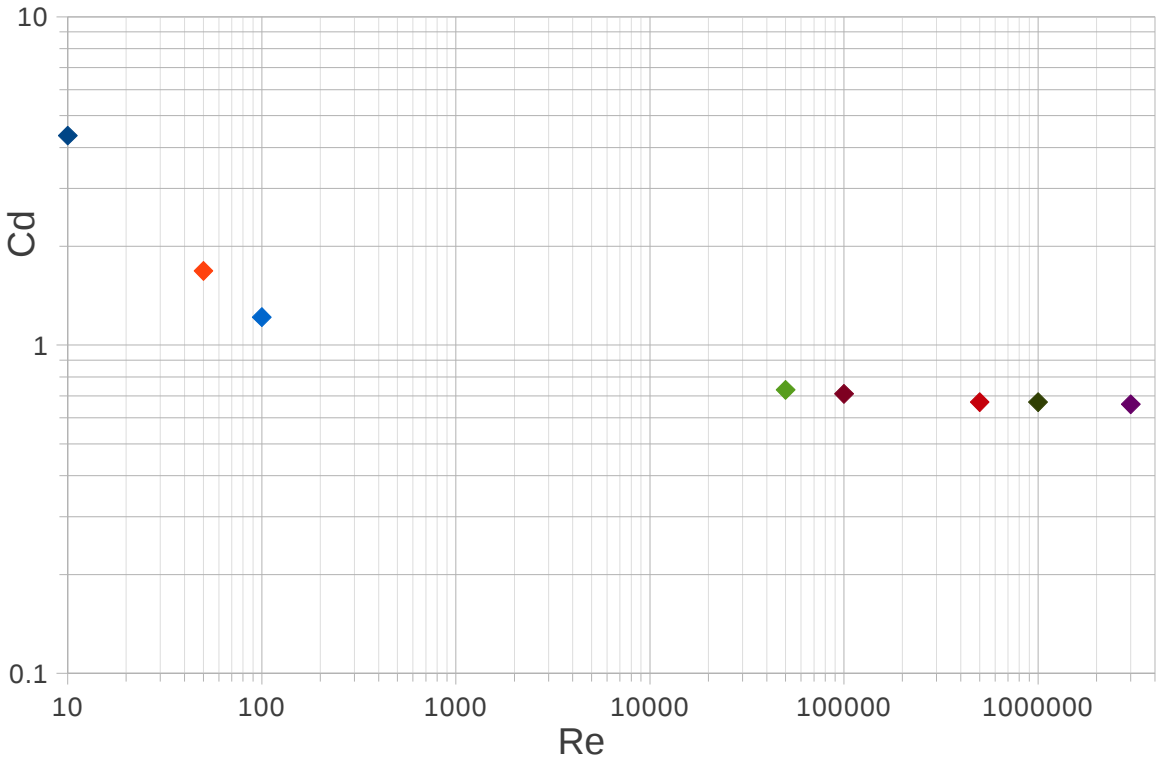
Re = 1 000 000
Cd = 0.67

Re = 3 000 000
Cd = 0.66



Pressure coefficient distribution (Re = 100 000)





Y+ distribution for Re = 500 000